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TECH 4272

Week 4 Deliverables

1. High Availability Setup Implement basic HA

The screenshot shows the Proxmox VE 9.1.1 interface. On the left, the 'Server View' sidebar lists the 'Datacenter (cloud-cluster)' and 'testhost' node. The 'testhost' node is expanded, showing a list of VMs (100-109, 200, 9000, 9001) and storage options. The 'Shell' option is selected, opening a terminal window. The terminal shows the user 'Donovan' logging in as 'root' and running 'pvecm status'. The output displays cluster information for 'cloud-cluster', including version 1, transport 'knet', and secure authentication 'on'. It also shows quorum information with a date of 'Mon Apr 20 10:22:39 2026', provider 'corosync_votequorum', 1 node, and quorate status 'Yes'. Finally, it shows votequorum information with 1 expected and total vote, and membership information for node '0x00000001' (192.168.0.230) with 1 vote.

```
Donovan@testhost:~$ su -
Password:
root@testhost:~# pvecm status
Cluster information
-----
Name:          cloud-cluster
Config Version: 1
Transport:     knet
Secure auth:   on

Quorum information
-----
Date:          Mon Apr 20 10:22:39 2026
Quorum provider: corosync_votequorum
Nodes:         1
Node ID:       0x00000001
Ring ID:       1.a
Quorate:       Yes

Votequorum information
-----
Expected votes: 1
Highest expected: 1
Total votes:    1
Quorum:         1
Flags:          Quorate

Membership information
-----
Nodeid      Votes Name
0x00000001  1 192.168.0.230 (local)
root@testhost:~#
```

The screenshot shows the Proxmox VE 9.1.1 interface with the 'HA' (High Availability) status page selected in the left sidebar. The 'Status' section shows the cluster is 'OK' and lists the nodes: 'master' (testhost, active, Mon Apr 20 11:00:15 2026) and 'lrm' (testhost, active, Mon Apr 20 11:00:09 2026). The 'Resources' section shows a table of VMs and their status.

ID	State	Node	Name	Max. Restart	Max. Reloc...	Failback	Description
vm:100	started	testhost	FOSSBilling	1	1	true	
vm:200	started	testhost	HAPProxy-LB	1	1	true	

2. Automated Scaling Create script to distribute VMs across Proxmox nodes

The screenshot displays the Proxmox Virtual Environment 9.1.1 web interface. On the left, the 'Server View' sidebar shows a tree structure under 'Datacenter (cloud-cluster)' with a 'testhost' node expanded, listing various VMs and storage. The main panel shows the configuration for 'Node testhost', with the 'Shell' option selected. A terminal window titled 'load_balancer.py' is open, displaying a Python script for automated scaling. The script defines a 'LoadBalancer' class with methods to find the least loaded node and distribute VMs based on available CPU and memory resources.

```
GNU nano 8.4 load_balancer.py
class LoadBalancer:
    def __init__(self, proxmox_api):
        self.proxmox = proxmox_api
    def get_least_loaded_node(self):
        """Find node with most available resources"""
        nodes = self.proxmox.nodes.get()
        best_node = None
        best_score = 0
        for node in nodes:
            status = self.proxmox.nodes(node['node']).status.get()
            # Calculate available resources score
            cpu_available = 1 - (status['cpu'] / status.get('maxcpu', 1))
            mem_available = (status['memory']['free'] / status['memory']['total'])
            score = (cpu_available + mem_available) / 2
            if score > best_score:
                best_score = score
                best_node = node['node']
        return best_node
    def distribute_vm(self, template_id, vm_config):
        """Create VM on least loaded node"""
        target_node = self.get_least_loaded_node()
        # Create VM on selected node
        # (Modify provisioner to use specific node)
        pass
```

At the bottom of the terminal window, a status bar shows keyboard shortcuts: ^G Help, ^C Write Out, ^F Where Is, ^R Cut, ^E Execute, ^O Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line.

3. Disaster Recovery Plan Create DR procedures

Disaster Recovery Plan

Backup Strategy

- Daily VM backups (retention: 7 days)
- Daily database backups (retention: 14 days)
- Weekly configuration backups (retention: 4 weeks)

Recovery Procedures



Single Node Failure

1. Identify failed node
2. If cluster enabled: VMs automatically migrate
3. If standalone: Restore VMs from backup to new node
4. Update DNS/routing if needed
5. Notify affected customers

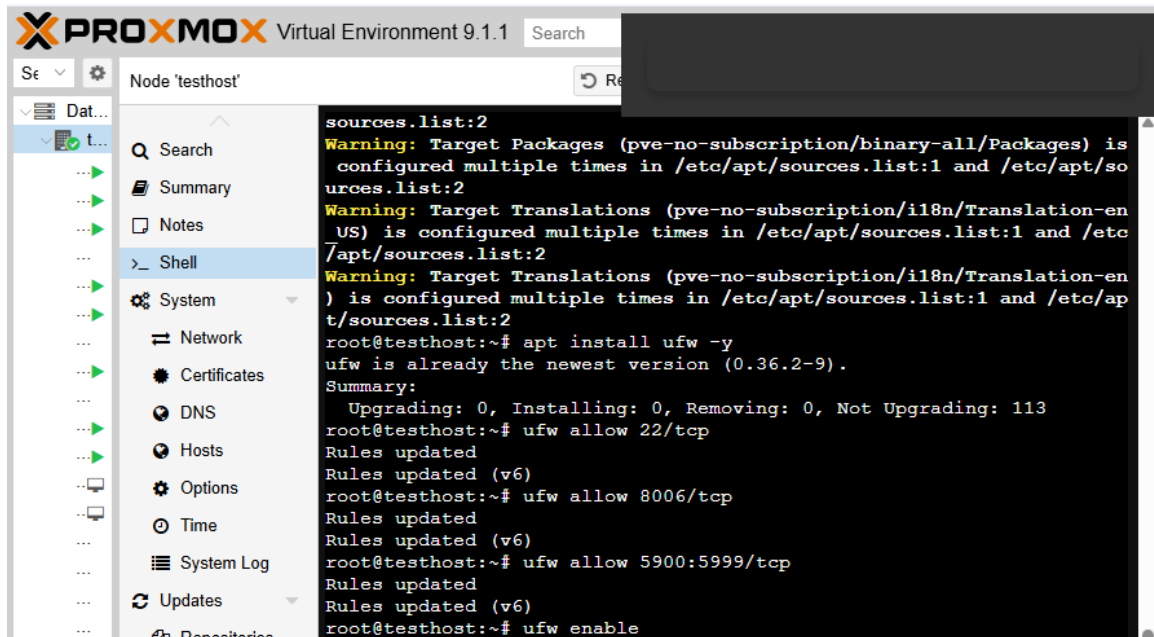
Database Corruption

1. Stop all services
2. Restore latest database backup
3. Verify data integrity
4. Restart services
5. Reconcile any lost transactions

Complete Site Failure

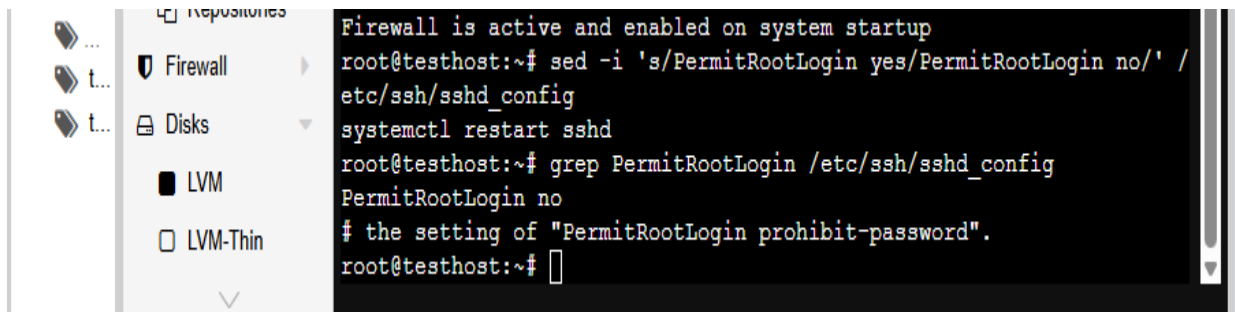
1. Provision new Proxmox infrastructure
2. Restore configuration backups
3. Restore database backup
4. Restore customer VMs from backups
5. Update DNS
6. Notify customers
7. Expected RTO: 4 hours

4. Security Hardening



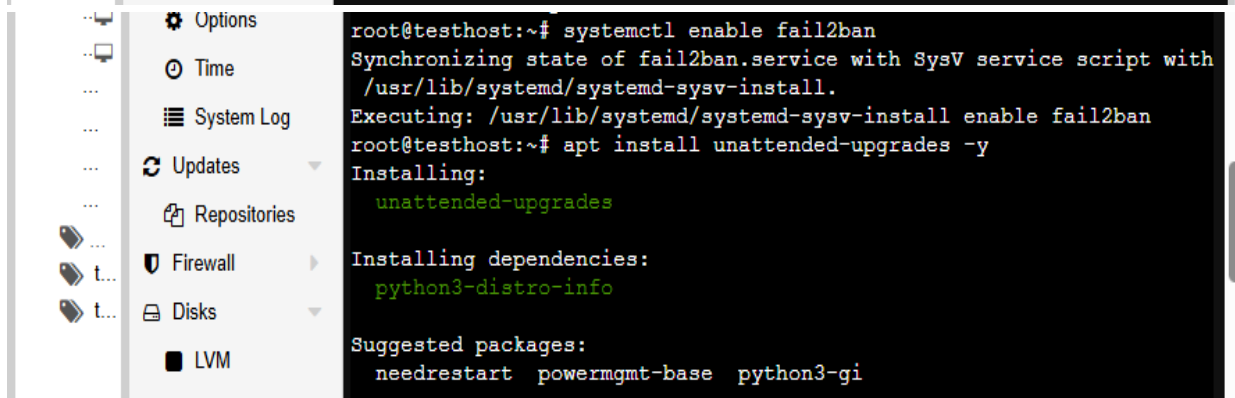
The screenshot shows the Proxmox VE 9.1.1 interface with the 'testhost' node selected. The left sidebar shows the 'Shell' option under the 'System' category. The terminal output shows the following commands and results:

```
sources.list:2
Warning: Target Packages (pve-no-subscription/binary-all/Packages) is
configured multiple times in /etc/apt/sources.list:1 and /etc/apt/so
urces.list:2
Warning: Target Translations (pve-no-subscription/i18n/Translation-en
US) is configured multiple times in /etc/apt/sources.list:1 and /etc
/apt/sources.list:2
Warning: Target Translations (pve-no-subscription/i18n/Translation-en
) is configured multiple times in /etc/apt/sources.list:1 and /etc/ap
t/sources.list:2
root@testhost:~# apt install ufw -y
ufw is already the newest version (0.36.2-9).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 113
root@testhost:~# ufw allow 22/tcp
Rules updated
Rules updated (v6)
root@testhost:~# ufw allow 8006/tcp
Rules updated
Rules updated (v6)
root@testhost:~# ufw allow 5900:5999/tcp
Rules updated
Rules updated (v6)
root@testhost:~# ufw enable
```



The screenshot shows the Proxmox VE 9.1.1 interface with the 'testhost' node selected. The left sidebar shows the 'Firewall' option under the 'System' category. The terminal output shows the following commands and results:

```
Firewall is active and enabled on system startup
root@testhost:~# sed -i 's/PermitRootLogin yes/PermitRootLogin no/' /
etc/ssh/sshd_config
systemctl restart sshd
root@testhost:~# grep PermitRootLogin /etc/ssh/sshd_config
PermitRootLogin no
# the setting of "PermitRootLogin prohibit-password".
root@testhost:~#
```



The screenshot shows the Proxmox VE 9.1.1 interface with the 'testhost' node selected. The left sidebar shows the 'Updates' option under the 'System' category. The terminal output shows the following commands and results:

```
root@testhost:~# systemctl enable fail2ban
Synchronizing state of fail2ban.service with SysV service script with
/usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
root@testhost:~# apt install unattended-upgrades -y
Installing:
  unattended-upgrades
Installing dependencies:
  python3-distro-info
Suggested packages:
  needrestart powermgmt-base python3-gi
```

